

PSMFlows: Decision Log

Running document — started August 2026

1 2026-08-05 — The fixed- u policy family is not goal-covering

Observations

1. **Reachability (exhaustive).** On pointmaze-medium task1, $d_a = 2$, so a 13×13 grid covers the whole latent box $[-3, 3]^2$. Rolling every grid latent, 32 dataset preimages, and 32 preimages of goal-reaching transitions as fixed- u policies (full 1000-step episodes): **0 of 233 ever reach the goal**; 75% never get closer than ~ 23 (maze span ~ 30); best is 1.32 at the saturated corner $u \approx [2, 2.5]$. [tools/latent_reachability.py]

Cube generalization (08-05, same tool): 2 of 128 candidates succeed (both prior draws, each in 1 of 2 episodes — $\sim 1\%$ of (latent, episode) pairs); goal-transition preimages again do no better than random draws. The family’s $\sim 1\text{--}2\%$ ceiling *quantitatively matches* the observed cube Stage-C GPI floor of 0.02–0.06: GPI was already extracting everything the family contains. *Antmaze (08-05):* 5 of 64 prior draws succeed, each in 1 of 2 episodes ($\sim 4\%$ of pairs). Family ceiling across the three envs: 0% / $\sim 1\%$ / $\sim 4\%$ — always floor-level, never usable.

2. **Latent coherence.** Along expert episodes, within-episode variance of the Stage-B preimage latents is $0.99\times$ the marginal variance (lag-1 autocorrelation 0.27, ~ 0 by lag 50); goal-reaching segments identical. A trajectory has no latent identity: the demonstrator’s route is *latent white noise*.
3. **Field structure.** Visualizing $a = G(xy, u)$ as a vector field per fixed u : typical latents give corridor-following fields with per-cell arbitrary direction choices $\Rightarrow$ circulation $\Rightarrow$ rollouts orbit near the start (path length 164, net displacement 1.4); saturated latents give near-constant headings (circular variance 0.01–0.03) that wall-slide diagonally and cannot turn. [tools/viz_fixed_u_field.py]
4. **Per-state coverage is fine.** At fixed cells, 512 latent draws span directions with circular variance 0.45–0.78: every direction remains available at every state. The deficiency is purely temporal.
5. **Downstream symptoms all follow.** ψ ’s TD target bootstraps $\psi(s', u', u')$ (continuation latent = index), so ψ faithfully represents a family whose members all share the same never-reach-goal value: $\psi^\top w$ is flat (spread $\sim 10\%$ of mean), the GPI argmax is noise and drifts to saturated corners, calibration reads predicted 85–212 vs. realized all-zero, and all eleven Stage-C runs score 0.0 (cube floors at 0.02–0.06).

Problem

Behavior cloning factorizes the data as (state $\mapsto$ conditional action distribution; $u \mapsto$ quantile within it). The observation lacks the demonstrator’s goal, so the goal-driven direction choice is forced into u *independently at each step*; nothing couples u across time. Rung-1 assumes u is a persistent policy index (“holding the noise fixed defines a family of policies”), but the family this actually defines contains only orbiters and constant headings — routes exist only as latent *sequences*. Stage C is not buggy; it succeeds at representing the wrong object. The support-constraint contribution (every considered action is a flow decode, $C = 1$) is untouched.

Fix

Drop “fixed noise = policy index.” Keep the frozen flow as the action-support constraint; move policy identity to a task vector w conditioning a *per-step* latent actor $u(s, w)$ (the Rung-3 amortized actor, built 08-04), and change the ψ backup to bootstrap the actor’s latent at $s' — \psi(s', \cdot, u(s', w))$ — i.e. policy improvement in latent space (PSM transposed to latents). Observation 4 says the per-step coverage this needs exists; every backup action remains a flow decode. Pending evidence: the running PSM baseline on pointmaze (RL-optimized family) is the control for whether improvement-in-the-family is the missing ingredient. Fallback if it also flatlines: trajectory-level latent in Stage A (skill-conditioned flow $G(s, u; w_{\text{traj}})$), at the cost of retooling Stages A–B.